$f(x) = x^2 - \cos x$

initial guess $x = 1$

solve & compare by Newton Raphson & one point-iteration method

Newton Raphson

$$x_{new} = x - \frac{f(x)}{f'(x)}$$

$f'(x) = 2x + \sin x$

	xx	xnew	ff
iter	x	xnew	f(xnew)
	1	0.8	0.01
	0.8		

```
for (i = 1 to 10)
{ xnew = x - f(x)/f'(x)
  xx.push(x)
  xnew.push(xnew)
  ff.push(f(xnew))
  x = xnew
}
```

one point-iteration method

$g(x) = \sqrt{\cos x}$

	xx	xnew	ff
iter	x	xnew = g(x)	f(x)
	1	0.8	0.01

```
for (i = 1 to 10)
{ xnew = g(x)
  xx.push(x)
  xnew.push(xnew)
  ff.push(f(xnew))
  x = xnew
}
```

Q Newton Raphson.